Chapter 3

Finite Differences

In this section, we introduce and analyze finite-difference methods. Throughout this section, we first discuss basic concepts using Model Problem 1 as an example, before proceeding to a somewhat more general description.

3.1 Grid

We approximate the derivates in Model Problem 1 with *finite differences*. First, we define a spatial *grid*. For now, such a grid is simply a set of equally spaced points in the interval $[0, 1]$:

$$\mathcal{G}_h := \{ih : i = 0, 1, \ldots, N;\ Nh = 1\}.$$

The setup of Model Problem 1 is such that we are given some initial temperature distribution, say at $t = 0$, and we are interested in the temperature distribution at some time $t = T > 0$. We need to solve the heat equation for $0 < t \leq T$, and so we define a set of points for the time interval $[0, T]$ as well:

$$\mathcal{G}_\tau := \{n\tau : n = 0, 1, \ldots, M;\ M\tau = T\}.$$

Finally we define a space-time grid covering the region of interest $\overline{\Omega} := [0, 1] \times [0, T]$ as the Cartesian product

$$\mathcal{G}_{h,\tau} := \mathcal{G}_h \times \mathcal{G}_\tau = \{(ih, n\tau) : ih \in \mathcal{G}_h,\ n\tau \in \mathcal{G}_\tau\}.$$

Figure 3.1 shows such a mesh. We use this set of discrete points to evaluate functions and their derivatives.

3.2 Difference Operators

Given that we evaluate functions at discrete grid points $(x_i, t^n) := (ih, n\tau)$, in order to approximate the PDE, we must find an approximation for the derivatives at these points. This is done via finite difference approximation, which can be easily derived using Taylor's series. To illustrate the approach, assume we are given a function $f : \mathcal{D} \subset \mathbb{R} \to \mathbb{R}$, which is k-times continuously differentiable. Then for some fixed $x \in \mathcal{D}$ and $x + h \in \mathcal{D}$ we can write the Taylor series

$$f(x + h) = \sum_{n=0}^{k-1} \frac{h^n}{n!} f^{(n)}(x) + \frac{h^k}{k!} f^{(k)}(\xi), \tag{3.1}$$

where the last term is the *remainder*, and ξ is some value between x and $x + h$.

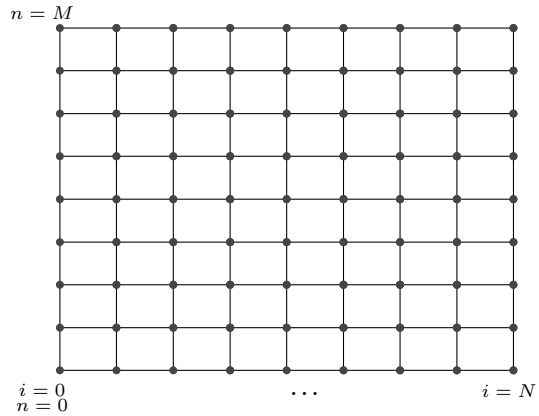

Figure 3.1: A two-dimensional grid for the space-time region $\overline{\Omega} := [0,1] \times [0,T]$. The grid is defined by the points $(ih, n\tau)$. For better visibility we have included lines of constant indices i and n as well.

Assume now that the exact solution u to our Model Problem 1 is at least four times continuously differentiable in space[1], and consider a Taylor Series approximation:

$$u(x_i \pm h, t^n) = u(x_i, t^n) \pm h u_x(x_i, t^n) + \frac{h^2}{2} u_{xx}(x_i, t^n) \pm \frac{h^3}{6} u_{xxx}(x_i, t^n) + \mathcal{O}(h^4).$$

From that there follows from a straightforward computation

$$u_{xx}(x_i, t^n) = \frac{u(x_i + h, t^n) - 2u(x_i, t^n) + u(x_i - h, t^n)}{h^2} + \mathcal{O}(h^2). \tag{3.2}$$

Similarly, assuming that u is at least two times continuously differentiable in time, we write the Taylor series in time as

$$u(x_i, t^n + \tau) = u(x_i, t^n) + \tau u_t(x_i, t^n) + \mathcal{O}(\tau^2).$$

We easily obtain

$$\frac{u(x_i, t^n + \tau) - u(x_i, t^n)}{\tau} = u_t(x_i, t^n) + \mathcal{O}(\tau). \tag{3.3}$$

In this section we discuss several schemes for model problem 1 to illustrate the choices one has when devising difference schemes. Subsequently we elaborate on the theory.

3.2.1 Explicit Scheme

Using the difference approximations (3.3) and (3.2), we obtain an approximation to the PDE (2.2). Let us denote the approximation to the solution at the grid point (x_i, t^n) by u_i^n. We choose u_i^n such that it satisfies

$$\frac{u_i^{n+1} - u_i^n}{\tau} = a \frac{u_{i-1}^n - 2u_i^n + u_{i+1}^n}{h^2}, \qquad i = 1, \ldots, N-1, \quad n = 0, \ldots M-1. \tag{3.4}$$

Thus, we replace (2.2), which gives a relation between space and time derivatives at each $(x,t) \in \Omega$ with a discrete approximation to be satisfied at all interior grid points. The scheme is completely defined if we combine this discretization with boundary conditions.

[1] We have already mentioned in the introduction that the solution to Model Problem 1 possesses infinitely many derivatives. So, at least for this problem, differentiability will not be a problem.

Scheme 1 (Explicit scheme for Model Problem 1) We write for $i = 1, \dots N-1$ and $n = 0, \dots M-1$:

$$u_i^{n+1} = \lambda u_{i-1}^n + (1-2\lambda)u_i^n + \lambda u_{i+1}^n. \tag{3.5}$$

Upon defining $\lambda := a\tau/h^2$, eq. (3.5) follows immediately from (3.4). Boundary conditions are descritized by setting

$$\begin{aligned} u_i^0 &= u_0(ih), & i &= 0, 1, \dots, N \\ u_0^n &= 0, & n &= 1, \dots, M \\ u_N^n &= 0, & n &= 1, \dots, M \end{aligned}$$

We say Scheme 1 is a *one-step explicit method*. It is called a one-step method, because only the previous time instance n is coupled to the time instance $n+1$. To say that a scheme is explicit means that each new value at $n+1$ is obtained directly by evaluating only known values at n. Figure 3.2 shows the *stencil* of the explicit scheme. The stencil is simply the grid points involved in the equation for u_i^{n+1}. (See eq. (3.5).)

For the explicit scheme discussed here, the stencil also has a direct interpretation as the *domain of dependence*. The solution at (x_i, t^n) depends only on those points at time level n which appear in the stencil. This finite domain of dependence is in stark contrast to the domain of dependence of the heat equation: As we have seen in the introduction (cf. Section 1.3.1), the solution at any point in the interior of the domain depends on all of the initial conditions.

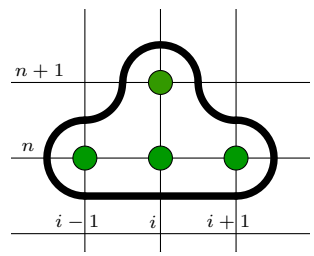

Figure 3.2: Stencil of the explicit scheme.

Note that non-homogeneous boundary conditions can be easily included. For example, if we are given Dirichlet boundary conditions $u(0,t) = g_1(t)$ and $u(1,t) = g_2(t)$ for some continuous functions g_1, g_2, we can set, for $n = 1, 2, \dots M$,

$$u_0^n = g_1(t^n), \tag{3.6}$$
$$u_N^n = g_2(t^n). \tag{3.7}$$

This means that we can sample the boundary conditions from the correct Dirichlet data. (Note that the initial conditions were handled in exactly the same manner.)

3.2.2 Notation

Before we proceed, we discuss some notation. The numerical approximation, generated by Scheme 1, appears to be defined only at the grid points $(ih, n\tau) \equiv (x_i, t^n)$. For fixed n and fixed grid size N, we may collect all the values of the numerical solution into a vector

$$\mathbf{u}^n = \begin{pmatrix} u_0^n \\ \vdots \\ u_N^n \end{pmatrix}. \tag{3.8}$$

The values for u_0 and u_N are fixed by the boundary conditions. In the case of Dirichlet boundary conditions one can, and we generally will, eliminate those entries. For example, upon re-defining the vector

$$\mathbf{u}^n = \begin{pmatrix} u_1^n \\ \vdots \\ u_{N-1}^n \end{pmatrix}, \tag{3.9}$$

Scheme 1, which has homogeneous boundary conditions, may be written

$$\mathbf{u}^{n+1} = A_{h,\tau}\mathbf{u}^n.$$

where

$$A_{h,\tau} = \begin{pmatrix} 1-2\lambda & \lambda & & & & \\ \lambda & 1-2\lambda & \lambda & & & \\ & \lambda & 1-2\lambda & \lambda & & \\ & & & \ddots & & \\ & & & \lambda & 1-2\lambda & \lambda \\ & & & & \lambda & 1-2\lambda \end{pmatrix}, \tag{3.10}$$

and all entries not explicitly shown are zero. Note that it is implied that $u_0 = u_N = 0$. The same scheme with the inhomogeneous Dirichlet conditions (3.6), (3.7) can be written

$$\mathbf{u}^{n+1} = A_{h,\tau}\mathbf{u}^n + \mathbf{f}^n.$$

where $\mathbf{u}$ and $A_{h,\tau}$ are as before, while

$$\mathbf{f}^n = \begin{pmatrix} \lambda g_1(t^n) \\ 0 \\ \vdots \\ 0 \\ \lambda g_2(t^n) \end{pmatrix}. \tag{3.11}$$

It is worth pointing out that the size of the matrix $A_{h,\tau}$ depends on the discretization parameter h. With our current practice of not including the boundary nodes in our solution vector the matrix is of size $(N-1)\times(N-1)$, where $N = 1/h$.

3.2.3 Implicit Methods

Instead of using a forward difference in time, as in (3.4), we can use a backward difference, and write

$$\frac{u_i^n - u_i^{n-1}}{\tau} = a\frac{u_{i-1}^n - 2u_i^n + u_{i+1}^n}{h^2}, \quad i = 1,\dots,N-1, \quad n = 1,\dots,M$$

or, after a simple index shift

$$\frac{u_i^{n+1} - u_i^n}{\tau} = a\frac{u_{i-1}^{n+1} - 2u_i^{n+1} + u_{i+1}^{n+1}}{h^2}, \quad i = 1,\dots,N-1, \quad n = 0,\dots,M-1.$$

Even if all the values at u_i^n are known, this cannot be solved for u_i^{n+1} by direct substitution, since all the unknown values at t^{n+1} are coupled; we call the scheme implicit. One needs to solve a linear system at each time step to get $\mathbf{u}_h^{n+1}$. Let

$$\lambda := \frac{a\tau}{h^2},$$

and define the implicit scheme

Scheme 2 (Implicit Scheme for Model Problem 1) For $n = 0,\dots,M-1$ solve:

$$-\lambda u_{i+1}^{n+1} + (1+2\lambda)u_i^{n+1} - \lambda u_{i-1}^{n+1} = u_i^n, \qquad i = 1,\dots N-1$$

with boundary conditions

$$\begin{aligned} u_i^0 &= u_0(ih), \qquad i = 0,1,\dots,N, \\ u_0^n &= 0, \qquad n = 1,\dots,M, \\ u_N^n &= 0, \qquad n = 1,\dots,M. \end{aligned}$$

The linear system with $N-1$ unknowns and $N-1$ equations that we must solve at each time step can be written in matrix form

$$A_{h,\tau}\mathbf{u}^{n+1} = \mathbf{u}^n,$$

where $\mathbf{u}$ is defined as in (3.9). The matrix $A_{h,\tau}$ is now defined as

$$A_{h,\tau} = \begin{pmatrix} 1+2\lambda & -\lambda & & & & \\ -\lambda & 1+2\lambda & -\lambda & & & \\ & -\lambda & 1+2\lambda & -\lambda & & \\ & & & \ddots & & \\ & & & -\lambda & 1+2\lambda & -\lambda \\ & & & & -\lambda & 1+2\lambda \end{pmatrix}. \tag{3.12}$$

This is a *tridiagonal matrix*. Appendix C provides a short overview of eigenvalues and eigenvectors for such matrices. By explicit computation, following the general case outlined in Appendix C, one can show that all the eigenvalues of $A_{h,\tau}$ are strictly positive. This implies that the matrix is *symmetric positive definite* (s.p.d.). It is thus non-singular and may be inverted. Furthermore, any s.p.d. matrix admits an LU-decomposition without pivoting, which simplifies solving linear systems. For this tridiagonal system, the cost for LU decomposition is $\mathcal{O}(N)$. (This will be covered in the tutorial.)

3.2.4 Combined Methods

Let $\Theta \in [0,1]$, and consider the difference approximation for the heat equation (2.2) given by

$$\frac{u_i^{n+1}-u_i^n}{\tau} = a\Theta\frac{u_{i-1}^{n+1}-2u_i^{n+1}+u_{i+1}^{n+1}}{h^2} + a(1-\Theta)\frac{u_{i-1}^n-2u_i^n+u_{i+1}^n}{h^2}.$$

This may be called a combined scheme in the sense that it offers a convex combination of the implicit and explicit discretization scheme, while the extreme values of $\Theta = 0$ and $\Theta = 1$ allow us to recover the purely explicit scheme of Section 3.2.1, and the purely implicit scheme of Section 3.2.3, respectively. Another special case is given for $\Theta = 1/2$, for which we can write the scheme

$$\frac{u_i^{n+1}-u_i^n}{\tau} = a\left(\frac{1}{2}\frac{u_{i-1}^{n+1}-2u_i^{n+1}+u_{i+1}^{n+1}}{h^2} + \frac{1}{2}\frac{u_{i-1}^n-2u_i^n+u_{i+1}^n}{h^2}\right).$$

Again defining $\lambda = a\tau/h^2$, we have the so-called *Crank-Nicolson* scheme:

Scheme 3 (Crank-Nicolson Scheme for Model Problem 1) For $n = 0,\dots,M-1$ solve:

$$-\frac{\lambda}{2}u_{i-1}^{n+1} + (1+\lambda)u_i^{n+1} - \frac{\lambda}{2}u_{i+1}^{n+1} = \frac{\lambda}{2}u_{i-1}^n + (1-\lambda)u_i^n + \frac{\lambda}{2}u_{i+1}^n, \qquad i = 1,\dots,N-1$$

with boundary conditions

$$\begin{aligned} u_i^0 &= u_0(ih), \qquad i = 0,1,\dots,N \\ u_0^n &= 0, \qquad n = 1,\dots,M \\ u_N^n &= 0, \qquad n = 1,\dots,M \end{aligned}$$

Scheme 3 can be written in matrix form as

$$A_{h,\tau}\mathbf{u}^{n+1} = B_{h,\tau}\mathbf{u}^n,$$

where

$$A_{h,\tau} = \begin{pmatrix} 1+\lambda & -\frac{\lambda}{2} & & & & \\ -\frac{\lambda}{2} & 1+\lambda & -\frac{\lambda}{2} & & & \\ & -\frac{\lambda}{2} & 1+\lambda & -\frac{\lambda}{2} & & \\ & & & \ddots & & \\ & & & -\frac{\lambda}{2} & 1+\lambda & -\frac{\lambda}{2} \\ & & & & -\frac{\lambda}{2} & 1+\lambda \end{pmatrix}, \tag{3.13}$$

$$B_{h,\tau} = \begin{pmatrix} 1-\lambda & \frac{\lambda}{2} & & & & \\ \frac{\lambda}{2} & 1-\lambda & \frac{\lambda}{2} & & & \\ & \frac{\lambda}{2} & 1-\lambda & \frac{\lambda}{2} & & \\ & & & \ddots & & \\ & & & \frac{\lambda}{2} & 1-\lambda & \frac{\lambda}{2} \\ & & & & \frac{\lambda}{2} & 1-\lambda \end{pmatrix}. \tag{3.14}$$

3.3 Interpretation

For the analysis of difference schemes it is often more convenient to *not* regard the numerical solution as a vector, but as a *grid function* by connecting the values at the grid nodes with piecewise affine linear functions (cf. Fig. 3.3). Clearly both representations are uniquely defined

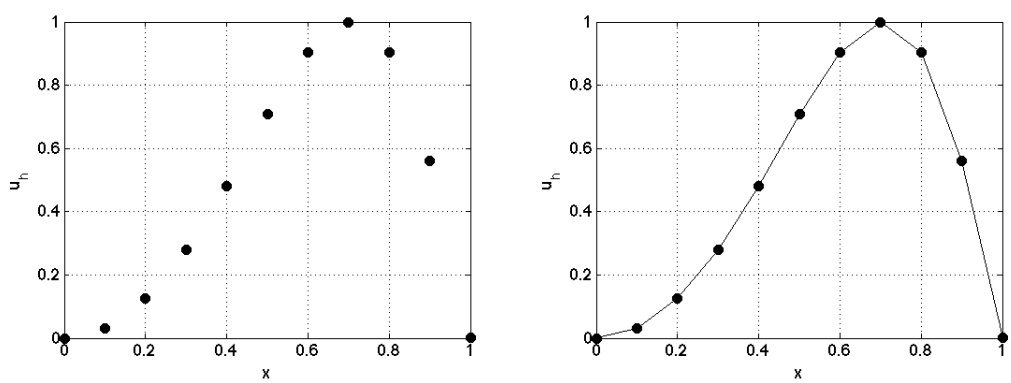

Figure 3.3: The numerical solution produced by a difference scheme may be interpreted as being defined only at the grid nodes (left), or as a piecewise linear continuous function (right).

by the values u_i^n on the grid. Higher order polynomial or trigonometric representation may be considered as well. In principle, *any* representation that is uniquely specified by the values at the nodes can be thought of as a representation of the numerical solution. However, the numerical analysis of higher order polynomial representations is nontrivial by itself, and we will, at least in the beginning, restrict ourselves to the viewpoints shown in Figure 3.3. Viewing the numerical solution as a function, defined for all x, often allows us to view both the solution to a general PDE problem, and the corresponding numerical solution as members of certain subspaces of the same Banach space V. (In the case of Model Problem 1, this would be the continuous functions that satisfy the boundary conditions, defined in (2.4).)

We may denote the grid function representing the numerical solution at time t^n as $u_h^n \in V$. It is quite apparent that at each time instance n, Scheme 1 merely transforms one grid function into another. We may interpret one step of the difference approximation as the application of an operator $E_\Delta(\tau) : V \to V$ providing the mapping $u_h^{n+1} = E_\Delta(\tau)u_h^n$. Recursively, we obtain[2]

[2]Note that the "$n+1$" in $E_\Delta(\tau)^{n+1}$ is an exponent, not a superscript.

$u_h^{n+1} = E_\Delta(\tau)^{n+1} u_h^0$. The matrix A defined in (3.10) is a proper representation of the operator E_Δ for a given (finite) grid size. It is easy to see that E_Δ is a linear operator.

We will generally assume that h and τ are not independent, but that instead there is a relation, say $h = h(\tau)$, such that $h \to 0$ if $\tau \to 0$. This justifies the notation $E_\Delta(\tau)$, i.e. characterizing the numerical scheme by a one-parameter family of operators.